INCH-POUND

A-A-50900A
15 January 1993
SUPERSEDING
A-A-50900
14 September 1983

COMMERCIAL ITEM DESCRIPTION

STERILIZER, SURGICAL INSTRUMENT AND DRESSING

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification, MIL-S-36642C (DSA-DM) which is cancelled

This Commercial Item Description covers a portable sterilizer suitable for sterilizing dental and surgical instruments and dressing.

Salient characteristics:

The unit shall be a portable sterilizer, electrically heated, automatically controlled, high speed, pressure, single shell, encased in a cabinet, suitable for sterilizing unwrapped, wrapped packed, and liquid products in hospital, laboratory, scientific and medical office environments.

The sterilizer shall consist of a cabinet, sterilizing chamber, water reservoir, electric heater, control, and valves, and shall have at least one easily removable tray and rack fabricated of corrosion-resisting metal.

Maximum overall dimensions and minimum chamber dimensions of the sterilizer shall be as follows:

Overall length:	24 inches
Overall width:	20 inches
Overall height:	21 inches
Chamber diameter:	7-7/8 inches
Chamber usable depth:	14 inches

AMSC N/A FSC 6530

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The sterilizer shall have the following operating controls, instruments, and visible signal devices mounted in the face panel for ease of reading and accessibility:

* A temperature controller/indicator which shall control and indicate the temperature between 212°F-270°F in the chamber, and shall start the timing phase of the cycle at set temperature.

* A cycle timer which shall control the period of time that the load will be exposed to steam at set temperature.

* A processing or condition light which shall indicate the chamber is being heated to operating temperature. It shall come on when the door is in the locked position and time is set on the cycle timer.

* A sterilize light which shall come on during a pressurized cycle when temperature has been attained.

* A light indicating a drying phase which shall come on when the door is opened and time is set on cycle timer for drying a load at low temperature.

* A reset button which shall restore operation after overheat switch has tripped to prevent overheating when water is low in chamber. As an alternate, the unit shall have a suitable fail-safe method to render the unit inoperable until the overheat condition has been eliminated.

Cycle selection shall be for at least wrapped goods, unwrapped instruments and liquids. Cycle phase timers shall be a minimum of 0-30 minutes for drying phase and 3-90 minutes for exposure phase, operator selectable with factory preset default settings as follows:

a. Instrument or soft goods (unwrapped instrument (flash) cycle) - 3 minutes exposure at 275°F and 15 minutes drying time.

b. Liquid cycle - 30 minutes exposure at 250°F with no drying time. The unit shall automatically control the return of the chamber to atmospheric pressure at a rate of approximately 4.5°F per minute to prevent liquid boilover.

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c. Wrapped goods cycle - 10 minutes exposure at 250°F, with adjustable dry time.

There shall be an audible signal for cycle complete. A visible indication of an overheat condition shall be displayed to notify the operator.

The sterilizer shall be fabricated of corrosion-resisting metal with the exception of minor parts such as knobs, handles, controls, insulation, etc. As an alternate, the outer cabinet may be constructed of a combination of high temperature polysulfane, heat resistant ABS or a suitable material, finished to resist corrosion.

The unit shall be capable of completing at least ten sterilization cycles, with load, on one filling of the reservoir. It shall further be designed to return the residual steam and condensate in the chamber to the reservoir following a sterilization cycle. The reservoir shall be completely enclosed within the cabinet. The fill opening for the reservoir shall be located on the top of the cabinet. The reservoir shall have a cover to adequately prevent inadvertent spillage of distilled water during pitch and roll movements when installed aboard naval vessels or other transport modes. A drain connection for emptying the reservoir shall be accessible from the exterior of the cabinet.

The operating temperature range shall be 250°F to 270°F. The sterilizer shall be capable of attaining and maintaining a temperature of 270°F for five minutes, and 250°F for 20 minutes. The sterilizer control system shall control chamber temperature to within plus 6°F minus 2°F of the set point.

The unit shall be constructed to satisfactorily withstand a chamber pressure of 60 psi and shall include a safety valve set at a pressure between 38 psi and 45 psi.

The sterilizer shall have a chamber door mechanism that will offer resistance to the user such that the door cannot be opened until chamber pressure is vented and reduced to at least 3 psig. As an alternate, it shall be necessary to manually activate the chamber exhaust vent, before the door latch can be released.

A three conductor line cord at least six feet in length, equipped with a three prong male plug shall be furnished.

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Average cycle time with 2 pounds of unwrapped instruments shall not exceed 28 minutes from a cold start and 20 minutes from a hot start.

The sterilizer shall comply with ASME Pressure Vessel Code, with UL Standard 544 and shall be listed by ETL, UL or other nationally recognized testing laboratories.

The maximum load size which the unit shall be capable of processing shall be:

a. Six pounds unwrapped instruments
b. Three pounds double wrapped instruments
c. Three 500 ML beakers 75 percent full
d. Linen pack which contains two towels, two sponges, and one tape sponge

The unit shall be designed to operate in altitude from sea level to 10,000 feet.

The sterilizer shall be furnished with two (2) complete sets of instructions for operation, maintenance and repair. Included shall be wiring diagrams, illustrations, and a list of all components with manufacturer's part number for other than common hardware.

Electrical requirements. The sterilizer shall be capable of operating on 120 volts, 50/60 hertz, AC power.

Workmanship. The sterilizer shall be rigidly constructed for strength and durability. All exposed edges, if not hemmed, beaded, or flanged, shall be smoothly finished to remove all sharp edges. The finished sterilizer shall be clean, well made, and free from any defect which affects its appearance or may impair its serviceability.

Unit. Each (EA). One sterilizer, as specified, constitutes one unit.

Quality Assurance Provisions.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections

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set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with an AQL of 2.5 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one sterilizer.

Sampling for tests. Sampling for tests, including dimensional test, shall be conducted in accordance with MIL-STD-105, with an AQL of 2.0 (percent defective) and an inspection level of II. The unit of product for test purposes shall be one sterilizer.

Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more test characteristics. Tests shall be in accordance with the latest edition of ANSI/ADA specification No. 59 for Portable Steam Sterilizers.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

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The contracting officer has the option of accepting or rejecting the product.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Regulatory requirements.

Federal Food, Drug and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered materials. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order.

Ordering data. Intermediate/exterior package quantities, labeling, and marking must be specified in the contract and/or order.

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NOTE: The following National Stock Number is covered by this document.

6530-01-128-9481

MILITARY INTERESTS:

Custodians:

Air Force - 03
Army - MD
Navy - MS

PREPARING ACTIVITY:

DoD-MB

Agent:

DLA - DM

CIVIL AGENCY COORDINATING ACTIVITIES:

FDA - MPQAS
USPHS
VA - OSS

Project Number: 6530-2199

Location: STER1BC.DOC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I. RECOMMEND A CHANGE:	1. DOCUMENT NUMBER A-A-50900A	2. DOCUMENT DATE (YYMMDD) 15 January 1993

3. DOCUMENT TITLE
STERILIZER, SURGICAL INSTRUMENT AND DRESSING

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DEFENSE PERSONNEL SUPPORT CENTER ATTN: DPSC-MSE	b. TELEPHONE (Include Area Code) (1) Commercial AC 215-737-2870	(2) AUTOVON AV 444-2870
c. ADDRESS (Include Zip Code) 2800 SO. 20TH STREET PHILADELPHIA, PA 19145	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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